Field	Description
	F1. Review alone changes no target state. F2. Failed or repeated resolution does not partially change the case.
Related Use Cases	UC-12 – Report Suspicious Recruitment Content UC-13 – Administer Platform Access and Job Visibility
Priority	Medium

1.6 Chapter Summary

This chapter defined CareerFit's actors, functional and non-functional requirements, and fourteen approved role-oriented use cases. The specifications distinguish actor goals from internal processing and keep CV–Job Matching, Job Recommendation, Applications, AutoFit eligibility, notification delivery, reporting, moderation, and actionable email behavior traceable to verified implementation.

CHAPTER 2. THEORETICAL BACKGROUND

2.1 Recruitment Information and CV–Job Description Matching

Recruitment matching is an information-retrieval and decision-support problem that compares different types of evidence about candidates and vacancies. A curriculum vitae usually describes education, work history, projects, technical skills, certifications, languages, and contact information. A job description lists responsibilities, required and preferred skills, seniority, location, employment conditions, and company context. Some data is structured, such as years of experience or location, while other data is free text and may use inconsistent terms. Therefore, exact database filters are useful for fixed conditions but are not enough to rank the overall fit between a candidate and a position.

The same information supports different user tasks. From the candidate perspective, the system retrieves and ranks jobs that fit a CV or desired profile. From the recruiter perspective, it ranks applicants or discovers potential candidates for a specific vacancy. These directions share document representations and similarity functions, but they are not identical business decisions. A relevant job recommendation does not imply that an application has been submitted, and a high candidate–job similarity does not imply that a recruiter should accept the candidate. The score is therefore evidence for prioritization rather than a final employment decision.

Recruitment data also has domain-specific ambiguity. A technology may be expressed by an abbreviation, a product name, or a broader skill family; job titles vary across companies; and the same term may have different importance at junior and senior levels. Open recruitment corpora are relatively uncommon because CVs contain personal information. The Djinni Recruitment Dataset illustrates both the scale of the matching problem and the value of anonymized, documented corpora, providing more than 150,000 jobs and 230,000 candidate profiles in English and Ukrainian [6]. CareerFit is narrower: it focuses on IT recruitment and uses explicit fields and normalized text so that matching behavior can be inspected within the project scope.

2.2 Text Representation and Preprocessing

2.2.1 Document Normalization

Before vectorization, text must be converted into a consistent form. A typical pipeline extracts machine-readable text, normalizes character encoding and letter case, separates or standardizes tokens, removes formatting noise, and may filter stop words or apply stemming or lemmatization. For recruitment documents, preprocessing must keep important technical tokens such as framework names, programming languages, versions, and abbreviations. Removing too much information can make a CV and a JD appear less related even when they share an important technology.

Bilingual data adds further complexity. Vietnamese contains diacritics and multi-syllable expressions separated by spaces, while English technical terminology frequently appears unchanged inside Vietnamese sentences. A practical pipeline must apply deterministic normalization to equivalent inputs and maintain a controlled vocabulary. It should also keep structured attributes—such as location, language, seniority, and salary mode—available for validation or filtering instead of forcing every business constraint into a single text vector.

Preprocessing quality directly affects every later score. Missing OCR text, malformed documents, extremely short descriptions, duplicated boilerplate, and contradictory structured fields can distort term frequencies. For this reason, validation belongs before scoring. Hard validation rejects inputs that cannot support meaningful processing, while soft validation allows processing but records warnings or suggestions. This distinction prevents the ranking engine from silently assigning apparently precise scores to unusable data.

2.2.2 Bag-of-Words Representation

In a bag-of-words representation, a document is represented by the terms it contains and their weights; word order is not modeled directly. If the vocabulary contains |V| terms, each document is mapped to a vector in R^|V|. This representation is sparse because a particular CV or JD contains only a small portion of the complete vocabulary. Its main advantages are computational simplicity, reproducibility, and the ability to inspect which terms contribute to similarity. Its main limitation is lexical dependence: synonyms and related skills do not match unless normalization or an explicit mapping connects them.

2.3 TF-IDF and Cosine Similarity

2.3.1 Term Frequency and Inverse Document Frequency

The vector-space model represents documents and queries as weighted term vectors and ranks documents according to their geometric relationship [1]. TF-IDF is a standard weighting family for this model [2]. Term frequency (TF) reflects how strongly a term occurs in a particular document. Inverse document frequency (IDF) reduces the influence of terms that occur in many documents and increases the relative influence of terms that distinguish a smaller subset of the corpus.

Let tf(t,d) denote the frequency of term t in document d, let N be the number of documents in the reference corpus, and let df(t) be the number of documents containing t. A basic formulation is:

$$tfidf(t,d) = tf(t,d) \times log(N / df(t)).$$

Implementations often use logarithmic TF scaling, smoothed IDF, or vector normalization to handle repeated terms and unseen values. The exact formulation changes numeric scores, so a thesis must document the implemented formula rather than referring to TF-IDF as if it were a single universal function. A controlled or static reference corpus can improve score stability because the IDF of a term does not change whenever a user adds one document; however, it may become less representative as technologies and hiring terminology evolve.

2.3.2 Cosine Similarity

Cosine similarity compares the direction of two vectors rather than their unnormalized magnitude. For vectors x and y, it is defined as:

$$cos(x,y) = (x \cdot y) / (||x||_2 \, ||y||_2).$$

For non-negative TF-IDF vectors, cosine similarity usually lies between 0 and 1. A larger value indicates a greater proportion of shared weighted terms. Length normalization is valuable for CV–JD comparison because a long CV should not receive a higher score merely because it contains more words. TF-IDF-weighted cosine distance has also been used as an efficient and reproducible document-alignment method [3].

Cosine similarity remains a lexical measure. If a JD uses “container orchestration” while a CV only uses “Kubernetes,” the vectors may not overlap unless the vocabulary or normalization layer establishes that relationship. Conversely, shared generic terms can increase similarity without proving that mandatory requirements are satisfied. CareerFit therefore treats similarity as one component of a broader workflow that also includes structured fields, validation, labels, reasons, user feedback, and policy conditions.

2.4 Matching, Recommendation, and Ranking

Matching and recommendation can be formalized as ranking tasks. Given a query representation q and a candidate set D, the system computes a relevance score s(q,d) for each $d \in D$ and orders the results by decreasing score. For candidate-to-job matching, q may be a CV vector and D the active job vectors. For recruiter-side discovery, q may be the selected JD and D the available candidate or CV vectors. For profile-based recommendation, q represents the candidate's desired role, skills, location, seniority, and other preferences rather than one uploaded CV.

This separation matters because a CV describes past experience, while a preference profile describes what the candidate wants. A candidate may have Java experience but look for a Go position, so CV-based matching and preference-based recommendation can produce different but valid lists. Keeping the workflows separate also makes evaluation clearer because the labels, candidate sets, and user goals of one task should not be assumed to fit the other task.

A raw similarity value is not automatically a calibrated probability of hiring success. Mapping it to a percentage or label is a presentation and business-rule decision, not a statistical guarantee. Thresholds such as Low, Medium, High, or Potential should be documented, tested at boundary values, and kept distinct from application status. Stable secondary sorting is also necessary when multiple items have equal scores so that repeated requests do not appear inconsistent.

Recent resume–job matching research increasingly uses dense encoders and contrastive learning. ConFit v2, for example, improves resume–job representations through hypothetical resumes and hard-negative mining [7]. Such approaches can capture semantic relations beyond exact terms, but they require training data, model governance, and an evaluation design appropriate to the domain. CareerFit intentionally begins with a transparent lexical baseline whose calculations can be reproduced and explained, while treating semantic or hybrid retrieval as future work rather than an implemented result.

2.5 Relevance Feedback and the Rocchio Method

2.5.1 Relevance Feedback

Relevance feedback uses judgments about retrieved items to modify a query or profile representation. Instead of assuming that the original text completely expresses the user's information need, the system observes which results are marked relevant or non-relevant. Explicit feedback is especially useful in recruitment because the importance of related skills may be known to the candidate or recruiter but omitted from the original CV, profile, or JD.